

Ajay Venkatesh

Brooklyn, NY

+1 516 585 9013 Email LinkedIn GitHub Portfolio

Education

New York University | Master of Science in Computer Engineering (GPA – 3.9/4)

Aug 2024 – May 2026

Experience

Campus Mesh

Jan 2025 – Present

AI Software Engineer

New York, US

- Developed and maintained scalable **Node.js microservices** powering real-time academic workflows for **10K+ users**, including study groups, messaging, notifications, and profile management.
- Built a personalized **recommendation engine** using collaborative filtering and behavioral analytics, increasing user engagement by **28%** and retention by **50%**.
- Authored **LLM-based retrieval pipelines** for CampusIQ (AI Assistant), improving contextual accuracy by **30%** through semantic retrieval and structured data indexing.

Amazon

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, US

- Built a high-throughput, event-driven incident triage system using AWS ECS Fargate Tasks, SQS, and SNS, decoupling message processing and reducing incident response latency by **50%** during peak traffic.
- Designed and implemented a collaborative-filtering **recommendation engine** using user interaction signals (course views, clicks, and engagement data), increasing platform engagement by 28% and user retention by 50%.
- Improved large-scale log analysis by implementing a Model Context Protocol (MCP) server for efficient retrieval, achieving a **40%** reduction in AWS Athena query time and compute costs.
- Developed LLM-based semantic retrieval pipelines for CampusIQ using BERT-based sentence-transformer embeddings (MiniLM) with FAISS vector search and PostgreSQL metadata indexing to enable hybrid retrieval and improve response relevance by 30%.

New York University (Novel AI Technologies, Inc.)

Aug 2024 – Present

AI Software Engineer

New York, US

- Architected an **NSF-funded (\$100K)** AI-powered health platform for **iOS** and **Android** ingesting real-time wearable telemetry; implemented **time-series anomaly detection** using **Isolation Forest** and **LSTM models**.
- Built **computer vision pipelines (YOLOv8, EfficientNet)** to process user-submitted food images integrated with **GPT-4 workflows** for nutritional estimation and dietary feedback.
- Developed an insurance analytics platform (**React, Node.js, REST APIs**) integrating **LLM-based document analysis (GPT-4 + LangChain)** for underwriting insights.
- Engineered ETL pipelines transforming **NoSQL** data into **PostgreSQL** with concurrency control; processed **250K+** records and powered **AWS QuickSight** dashboards.

PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Software Engineer

Bangalore, India

- Designed and implemented a **distributed backend system** on **Microsoft Azure** integrating with **Microsoft Dynamics** and supporting ML inference pipelines for healthcare analytics.
- Led **code reviews** and mentored interns on software design and **test-driven development**, reducing production bugs by **25%**.
- Built an **NLP-powered OCR pipeline** using **LayoutLM** and **Azure Form Recognizer** to automate data extraction from medical documents, reducing processing time by **75%**.
- Optimized data ingestion using **parallel processing** for **100K+ Excel records**, reducing execution time from **5 hours** to **1 hour**.
- Developed **Power BI dashboards** integrating multiple enterprise databases, improving reporting efficiency by **40%**.

Projects

Flight Delay Prediction | PySpark, HDFS, SQL, Machine Learning

- Built a scalable PySpark ML pipeline on HDFS to process and engineer features from over 7M flight records using SQL-based aggregation and anomaly detection.
- Performed extensive EDA to identify delay patterns and key predictors; trained and optimized Random Forest and Gradient Boosted Tree models to achieve 85%+ accuracy.
- Converted model outputs into actionable insights, enabling data-driven scheduling and operational planning for delay reduction.

Skills

Programming Languages: C#, Python, Java, C++, JavaScript, SQL

AI/ML: Neural Networks, Classification, Pattern Recognition, Data Extraction, RAG, MLOps, AI Solution Deployment, Responsible AI, Model Monitoring, Model Validation

Web Development: REST API, GraphQL API, Microservices, React, Node.js, Express.js

Database: Dataverse, RDBMS (MySQL, PostgreSQL), NoSQL (MongoDB, JSON)

Cloud: Azure (AI Services, Azure AI Foundry, Cognitive Services, Logic Apps, Function Apps, Storage Blobs, Security Data Lakes), AWS (CDK, SDK, EC2, S3, Lambda, IAM, Q, SageMaker)

Automation Platforms: Power Automate, Microsoft Copilot Studio, Azure AI Studio

Tools: Git, Docker, Jupyter, Visual Studio, Postman, Power BI, CI/CD Pipelines

Methodology: Agile, Scrum, SDLC Governance, Test-Driven Development (TDD)

Certifications & Publications

AWS Cloud Practitioner

Stanford Machine Learning (Coursera)

Power BI Data Analyst Associate

Azure Data Fundamentals

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques (IEEE)

An Aquila-optimized SVM classifier for Diabetes prediction (IEEE)